GRBAS Trainer

User guide for perceptual training

Concept and clinical direction: Dr. Patricia G. Farías, Speech-Language Pathologist — Vocology Specialist, Buenos Aires, Argentina
Built with the assistance of Claude (Anthropic)

An auditory-perceptual calibration tool built on 296 voices from the Perceptual Voice Qualities Database (PVQD)

1. What this tool is

GRBAS Trainer is a training app for calibrating auditory-perceptual voice assessment with the GRBAS scale (Hirano, 1981: Grade, Roughness, Breathiness, Asthenia, Strain). It lets you rate each of 296 voices, compare your rating with the consensus of expert raters, and track your progress during training.

The voices come from the Perceptual Voice Qualities Database (PVQD), a public dataset created by Patrick R. Walden (St. John's University) for educational purposes: 296 voices rated by 19 experienced clinicians, with good inter- and intra-rater reliability.

2. Before you start

Important — audio is not included: this tool contains the ratings and data for all 296 voices, but not the sound files themselves. To be able to listen to the voices, you need to download the 296 .wav files separately, directly from the official source:

Perceptual Voice Qualities Database (PVQD) — Mendeley Data:
https://data.mendeley.com/datasets/9dz247gnyb/3

- On that page you'll find a button to download all the audio files (.wav format), along with the original rating spreadsheets. Access is free, no account required, under a CC BY 4.0 license.
- Save the 296 files in a single folder on your computer. Each file is named with the same participant ID used by the app (e.g. BL01.wav, SJ2003.wav, NYU1002.wav), so it's easy to find the one matching whichever voice you're rating.
- Downloading all 296 files may take a few minutes depending on your connection — it's a one-time step, you won't need to repeat it every time you train.

Recommended browser: use Chrome or Edge. Firefox blocks saving progress for locally-opened files as a security measure, so your ratings would not persist between sessions.

- Keep the .html file in a fixed folder from the start: saved progress is tied to that exact file location. If you later move or rename the file, your previous ratings won't show up (the browser treats it as a different file).
- Open the .html file by double-clicking it: it opens in your browser. No installation is required, and no internet connection either, except for the Dashboard charts to load.
- Your ratings are saved automatically in your browser and persist between sessions: you can close and reopen the file without losing your progress.
- Each person who opens the file on their own computer has their own rating history; ratings are not automatically shared between users.

- If you have the voice audio files (WAV/MP3), keep them accessible in a folder. The app does not open them automatically for browser security reasons: they must be loaded manually, one voice at a time.

3. "Voices" tab: the full list

This is the starting screen. It shows the 296 voices with the following information in each row:

- ID: the voice identifier (e.g. BL01, SJ2003, NYU1002).
- Sex and Age of the speaker — always visible, they don't affect the perceptual rating.
- Grade (consensus) and Dx (diagnosis) — hidden by default (see "Blind mode" below).
- Status: Pending or Rated, depending on whether you've already rated that voice.
- My agreement: once rated, how many of the 5 parameters matched the consensus exactly (e.g. 4/5).

Tools available on this tab:

- Search by ID, to jump straight to a specific voice.
- Filters: All / Pending / Rated.
- Sort by ID, Age, Biggest mismatch (useful to review the cases where you were furthest off first), or Most recently rated.
- Double-clicking any row (or the "Rate →" button) opens that voice in the Rate tab.

Blind mode

The "🙈 Blind mode: ON" button, enabled by default, hides the consensus Grade and Diagnosis for any voice you haven't rated yet. This avoids seeing the "answer" before listening and deciding, which would bias perception. Both fields are automatically revealed as soon as you save your rating for that voice. It can be turned off at any time to review the full list with the answers visible, for example at the end of a case conference.

4. "Rate" tab: how to score a voice

This is where the main work happens. The suggested procedure for each voice:

1. Check the basic data (ID, sex, age) shown in the left panel.
2. If you have the corresponding audio file, click "📂 Load audio" and select it from your computer. It plays back with the browser's standard controls.
3. Listen to the voice as many times as you need before deciding.
4. For each of the 5 parameters (G, R, B, A, S), pick a level from 0 to 3 by clicking the corresponding button.
5. Once all 5 parameters are set, the "Save rating" button becomes enabled. Click it to save.
6. The feedback panel appears immediately, comparing your rating against the consensus.

What to look at in the feedback

- For each parameter, your value vs. the consensus is shown, along with the difference (Δ). Δ = 0 is an exact match; a positive Δ means you rated it more severe than the consensus; a negative Δ, milder.
- The color of each cell indicates the size of the disagreement: exact match, a 1-point difference, or a 2-or-more-point difference (the latter is worth reviewing or discussing).

- Below that, the voice's diagnosis (now revealed) and your agreement with each individual PVQD rater are shown — useful for seeing whether your rating resembles one particular rater more closely.

The "Delete my rating" button removes your saved score for that voice, in case you made a mistake or want to try again without the previous information visible.

5. "Dashboard" tab: your training statistics

It updates automatically as you rate voices. No manual action is needed to see it.

- Top cards: number of voices rated, overall % exact match, mean absolute error (MAE), and average weighted kappa.
- "Statistics by parameter" table: the same type of metrics, broken down parameter by parameter (G, R, B, A, S) to identify where you perform better or worse.
- Bar chart: visually compares MAE and weighted kappa across the 5 parameters.
- Evolution chart: shows your % exact match in a rolling window of the last 10 voices rated, in chronological order — lets you see whether you're improving as you train.
- Export: you can download your complete ratings as JSON, or a row-by-row comparison against the consensus as CSV, for further analysis or to share with an instructor.

6. The GRBAS scale: quick reference

Severity levels (the same for all 5 parameters):

Level	Label	What it means
0	Normal	No perceived deviation in that parameter.
1	Mild	A slight deviation; when in doubt, score 1 rather than 0.
2	Moderate	A clearly perceptible deviation.
3	Severe	An extreme deviation.

The five parameters, and what to listen for in each:

Letter	Parameter	What to listen for
G	Grade	Overall impression of dysphonia: general severity, summarizing everything else.
R	Roughness	Rough, irregular quality; irregular vocal fold vibration.
B	Breathiness	Audible air escape due to incomplete glottal closure.
A	Asthenia	Weakness, lack of projection or energy in the voice.
S	Strain	Excessive effort; an impression of laryngeal hyperfunction.

7. Interpreting the Dashboard statistics

Statistic	How to interpret it
% exact	Percentage of ratings that match the consensus exactly. The closer to 100%, the better.
% ±1	Percentage of ratings within one scale point of the consensus (e.g. scoring 2 when the consensus is 1 or 3).
MAE (Mean Absolute Error)	The average absolute difference between your rating and the consensus. 0 is perfect; above 0.5 suggests a systematic discrepancy in that parameter.
Weighted kappa	Measures your agreement with the consensus, correcting for chance and penalizing larger differences more heavily. Ranges from -1 to 1: above 0.6 is substantial agreement; above 0.8, almost perfect.
Best rater agreement	Of the individual PVQD raters, which one's scoring pattern most resembles yours for that parameter.

Rough guide for weighted kappa: below 0.2 (slight/poor), 0.2–0.4 (fair), 0.4–0.6 (moderate), 0.6–0.8 (substantial), above 0.8 (almost perfect).

8. Training recommendations

- Work in short sessions (15–25 voices) rather than attempting all 296 at once; auditory fatigue affects perceptual quality.
- Periodically use the "Biggest mismatch" sort to specifically review the cases where you were furthest from the consensus — that's where the most valuable learning happens.
- Pay special attention to Roughness (R) and Asthenia/Strain (A/S): the literature shows these are generally the parameters with the lowest agreement even among expert raters, so a higher MAE there isn't necessarily an individual training problem.
- Check the evolution chart every so many sessions to confirm your learning curve is improving, not just your total number of voices rated.
- For group case-conference use: it's best if a single person controls the app and shares their screen, letting the group deliberate out loud before entering the agreed-upon score.

9. Sources and references

Data: Walden, P.R. (2020). Perceptual Voice Qualities Database (PVQD), Mendeley Data, V3. DOI: 10.17632/9dz247gnyb.3 — CC BY 4.0 License.

Paper: Walden, P.R. (2022). Perceptual Voice Qualities Database (PVQD): Database Characteristics. Journal of Voice, 36(6), 875.e15–875.e23. DOI: 10.1016/j.jvoice.2020.10.001.

Scale: Hirano, M. (1981). Clinical Examination of Voice (Disorders of Human Communication, vol. 5). Wien: Springer-Verlag.